Worked solutions to selected exercises from
Richard Hammack, *Book of Proof, edition 2.2*

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1 Sets

1.1 Introduction to Sets

A. Write each of the following sets by listing their elements between braces.

2. $\{3x + 2 : x \in \mathbb{Z}\}$

$\{..., -7, -4, -1, 2, 5, 8, 11, ...\}$

4. $\{x \in \mathbb{N} : -2 < x \leq 7\}$

$\{1, 2, 3, 4, 5, 6, 7\}$

6. $\{x \in \mathbb{R} : x^2 = 9\}$

$\{-3, 3\}$

8. $\{x \in \mathbb{R} : x^3 + 5x^2 = -6x\}$

$\{-3, -2, 0\}$

10. $\{x \in \mathbb{R} : \cos x = 1\}$

$\{2\pi x : x \in \mathbb{Z}\}$

12. $\{x \in \mathbb{Z} : |x| < 5\}$

$\{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$

14. $\{5x : x \in \mathbb{Z}, |2x| \leq 8\}$

$\{-20, -15, -10, -5, 0, 5, 10, 15, 20\}$

16. $\{6a + 2b : a, b \in \mathbb{Z}\}$

$\{2k : k \in \mathbb{Z}\} = \{..., -4, -2, 0, 2, 4, ...\}$

B. Write each of the following sets in set-builder notation.

18. $\{0, 4, 16, 36, 64, 100, ...\}$

$\{4k^2 : k \in \mathbb{Z}\}$

20. $\{..., -8, -3, 2, 7, 12, 17, ...\}$

$\{7 - 5x : x \in \mathbb{Z}\}$

22. $\{3, 6, 11, 18, 27, 38, ...\}$

$\{n^2 + 2 : n \in \mathbb{N}\}$

24. $\{-4, -3, -2, -1, 0, 1, 2\}$

$\{x \in \mathbb{Z} : -4 \leq x \leq 2\}$

26. $\{..., \frac{1}{27}, \frac{1}{9}, \frac{1}{3}, 1, 3, 9, 27, ...\}$

$\{3^x : x \in \mathbb{Z}\}$

28. $\{..., -\frac{3}{2}, -\frac{3}{4}, 0, \frac{3}{4}, \frac{3}{2}, \frac{9}{4}, 3, \frac{15}{4}, \frac{9}{2}, ...\}$

$\{\frac{3}{4}x - \frac{3}{4} : x \in \mathbb{Z}\}$

C. Find the following cardinalities.

30. $|\{\{1, 4\}, a, b, \{\{3, 4\}\}, \{\emptyset\}\}|$

5

32. $|\{\{\{1, 4\}, a, b, \{\{3, 4\}\}, \emptyset\}\}|$

1

34. $|\{x \in \mathbb{N} : |x| < 10\}|$

9

36. $|\{x \in \mathbb{N} : x^2 < 10\}|$

3

38. $|\{x \in \mathbb{N} : 5x \leq 20\}|$

4

1.2 The Cartesian Product

A. Write out the indicated sets by listing their elements between braces.

2. Suppose $A = \{\pi, e, 0\}$ and $B = \{0, 1\}$.

a) $A \times B = \{(\pi, 0), (\pi, 1), (e, 0), (e, 1), (0, 0), (0, 1)\}$

b) $B \times A = \{(0, \pi), (0, e), (0, 0), (1, \pi), (1, e), (1, 0)\}$

c) $A \times A = \{(\pi, \pi), (\pi, e), (\pi, 0), (e, \pi), (e, e), (e, 0), (0, \pi), (0, e), (0, 0)\}$

d) $B \times B = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$

e) $A \times \emptyset = \emptyset$

f) $(A \times B) \times B = \{$
$((\pi, 0), 0), ((\pi, 1), 0), ((e, 0), 0), ((e, 1), 0),$
$((0, 0), 0), ((0, 1), 0), ((\pi, 0), 1), ((\pi, 1), 1),$
$((e, 0), 1), ((e, 1), 1), ((0, 0), 1), ((0, 1), 1)$
$\}$

g) $A \times (B \times B) = \{$
$(\pi, (0, 0)), (\pi, (0, 1)), (\pi, (1, 0)), (\pi, (1, 1)),$
$(e, (0, 0)), (e, (0, 1)), (e, (1, 0)), (e, (1, 1)),$
$(0, (0, 0)), (0, (0, 1)), (0, (1, 0)), (0, (1, 1))$
$\}$

h) $A \times B \times B = \{$
$(\pi, 0, 0), (\pi, 0, 1), (\pi, 1, 0), (\pi, 1, 1),$
$(e, 0, 0), (e, 0, 1), (e, 1, 0), (e, 1, 1),$
$(0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1)$
$\}$

4. $\{n \in \mathbb{Z} : 2 < n < 5\} \times \{n \in \mathbb{Z} : |n| = 5\}$

$\{(3,-5),(3,5),(4,-5),(4,5)\}$

6. $\{x \in \mathbb{R} : x^2 = x\} \times \{x \in \mathbb{N} : x^2 = x\}$

$\{(0,1),(1,1)\}$

8. $\{0,1\}^4$

$\{$
$(0,0,0,0),(0,0,0,1),(0,0,1,0),(0,0,1,1),$
$(0,1,0,0),(0,1,0,1),(0,1,1,0),(0,1,1,1),$
$(1,0,0,0),(1,0,0,1),(1,0,1,0),(1,0,1,1),$
$(1,1,0,0),(1,1,0,1),(1,1,1,0),(1,1,1,1)$
$\}$

1.3 Subsets

A. List all the subsets of the following sets.

2. $\{1,2,\emptyset\}$

$\{\},\{1\},\{2\},\{\emptyset\},\{1,2\},\{1,\emptyset\},\{2,\emptyset\},\{1,2,\emptyset\}$

4. $\emptyset$

$\{\}$

6. $\{\mathbb{R},\mathbb{Q},\mathbb{N}\}$

$\{\},\{\mathbb{R}\},\{\mathbb{Q}\},\{\mathbb{N}\},\{\mathbb{R},\mathbb{Q}\},\{\mathbb{R},\mathbb{N}\},\{\mathbb{Q},\mathbb{N}\},\{\mathbb{R},\mathbb{Q},\mathbb{N}\}$

8. $\{\{0,1\},\{0,1,\{2\}\},\{0\}\}$

$\{\},\{\{0,1\}\},\{\{0,1,\{2\}\}\},\{\{0\}\},$
$\{\{0,1\},\{0,1,\{2\}\}\},\{\{0,1\},\{0\}\},\{\{0,1,\{2\}\},\{0\}\},\{\{0,1\},\{0,1,\{2\}\},\{0\}\}$

B. Write out the following sets by listing their elements between braces.

10. $\{X \subseteq \mathbb{N} : |X| \leq 1\}$

$\{\emptyset,\{1\},\{2\},\{3\},...\}$

12. $\{X : X \subseteq \{3,2,a\}\ and\ |X| = 1\}$

$\{3\}, \{2\}, \{a\}$

C. Decide if the following statements are true or false.

14. $\mathbb{R}^2 \subseteq \mathbb{R}^3$

False.

16. $\{(x,y) : x^2 - x = 0\} \subseteq \{(x,y) : x - 1 = 0\}$

False.

1.4 Power Sets

A. Find the indicated sets.

2. $\mathcal{P}(\{1,2,3,4\})$

$\{$
$\emptyset, \{1\}, \{2\}, \{3\}, \{4\},$
$\{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\}, \{3,4\},$
$\{1,2,3\}, \{1,2,4\}, \{1,3,4\}, \{2,3,4\},$
$\{1,2,3,4\}$
$\}$

4. $\mathcal{P}(\{\mathbb{R}, \mathbb{Q}\})$

$\{\emptyset, \{\mathbb{R}\}, \{\mathbb{Q}\}, \{\mathbb{R}, \mathbb{Q}\}\}$

6. $\mathcal{P}(\{1,2\}) \times \mathcal{P}(\{3\})$

$\{$
$(\emptyset, \emptyset), (\emptyset, \{3\}),$
$(\{1\}, \emptyset), (\{1\}, \{3\}),$
$(\{2\}, \emptyset), (\{2\}, \{3\}),$
$(\{1,2\}, \emptyset), (\{1,2\}, \{3\})$
$\}$

8. $\mathcal{P}(\{1,2\} \times \{3\})$

$\{\emptyset, \{(1,3)\}, \{(2,3)\}, \{(1,3),(2,3)\}\}$

10. $\{X \in \mathcal{P}(\{1,2,3\}) : |X| \leq 1\}$

$\{\emptyset, \{1\}, \{2\}, \{3\}\}$

12. $\{X \in \mathcal{P}(\{1,2,3\}) : 2 \in X\}$

$\{\{2\}, \{1,2\}, \{2,3\}, \{1,2,3\}\}$

B. Suppose that $|A| = m$ and $|B| = n$. Find the following cardinalities.

14. $|\mathcal{P}(\mathcal{P}(A))|$

2^{2^m}

16. $|\mathcal{P}(A) \times \mathcal{P}(B)|$

2^{m+n}

18. $|\mathcal{P}(A \times \mathcal{P}(B))|$

$2^{m \cdot 2^n}$

20. $|\{X \subseteq \mathcal{P}(A) : |X| \leq 1\}|$

$2^m + 1$

1.5 Union, Intersection, Difference

2. Suppose $A = \{0,2,4,6,8\}$, $B = \{1,3,5,7\}$ and $C = \{2,8,4\}$. Find:

a) $A \cup B = \{0,1,2,3,4,5,6,7,8\}$

b) $A \cap B = \emptyset$

c) $A - B = A$

d) $A - C = \{0,6\}$

e) $B - A = B$

f) $A \cap C = C$

g) $B \cap C = \emptyset$

h) $B \cup C = \{1,2,3,4,5,7,8\}$

i) $C - B = C$

4. Suppose $A = \{b, c, d\}$ and $B = \{a, b\}$. Find:

a) $(A \times B) \cap (B \times B) = \{(b, a), (b, b)\}$

b) $(A \times B) \cup (B \times B) = \{(b, a), (b, b), (c, a), (c, b), (d, a), (d, b), (a, a), (a, b)\}$

c) $(A \times B) - (B \times B) = \{(c, a), (c, b), (d, a), (d, b)\}$

d) $(A \cap B) \times A = \{(b, b), (b, c), (b, d)\}$

e) $(A \times B) \cap B = \emptyset$

f) $\mathcal{P}(A) \cap \mathcal{P}(B) = \{\emptyset, \{b\}\}$

g) $\mathcal{P}(A) - \mathcal{P}(B) = \{\{c\}, \{d\}, \{b, c\}, \{b, d\}, \{c, d\}, \{b, c, d\}\}$

h) $\mathcal{P}(A \cap B) = \{\emptyset, \{b\}\}$

i) $\mathcal{P}(A) \times \mathcal{P}(B) = \{$
$(\emptyset, \emptyset), (\emptyset, \{a\}), (\emptyset, \{b\}), (\emptyset, \{a, b\}),$
$(\{b\}, \emptyset), (\{b\}, \{a\}), (\{b\}, \{b\}), (\{b\}, \{a, b\}),$
$(\{c\}, \emptyset), (\{c\}, \{a\}), (\{c\}, \{b\}), (\{c\}, \{a, b\}),$
$(\{d\}, \emptyset), (\{d\}, \{a\}), (\{d\}, \{b\}), (\{d\}, \{a, b\}),$
$(\{b, c\}, \emptyset), (\{b, c\}, \{a\}), (\{b, c\}, \{b\}), (\{b, c\}, \{a, b\}),$
$(\{b, d\}, \emptyset), (\{b, d\}, \{a\}), (\{b, d\}, \{b\}), (\{b, d\}, \{a, b\}),$
$(\{c, d\}, \emptyset), (\{c, d\}, \{a\}), (\{c, d\}, \{b\}), (\{c, d\}, \{a, b\}),$
$(\{b, c, d\}, \emptyset), (\{b, c, d\}, \{a\}), (\{b, c, d\}, \{b\}), (\{b, c, d\}, \{a, b\})$
$\}$

10. Do you think the statement $(\mathbb{R} - \mathbb{Z}) \times \mathbb{N} = (\mathbb{R} \times \mathbb{N}) - (\mathbb{Z} \times \mathbb{N})$ is true, or false? Justify.

True. An ordered pair (x, n) lies in $(\mathbb{R} \times \mathbb{N}) - (\mathbb{Z} \times \mathbb{N})$ exactly when $(x, n) \in (\mathbb{R} \times \mathbb{N})$ but $(x, n) \notin (\mathbb{Z} \times \mathbb{N})$, that is, $x \in \mathbb{R}$ and $n \in \mathbb{N}$ but not both $x \in \mathbb{Z}$ and $n \in \mathbb{N}$. Since in both cases $n \in \mathbb{N}$, the second premise collapses to $x \notin \mathbb{Z}$, hence, the ordered pair satisfies $x \in (\mathbb{R} - \mathbb{Z})$ and $n \in \mathbb{N}$, that is, $(x, n) \in (\mathbb{R} - \mathbb{Z}) \times \mathbb{N}$, therefore, both sets have exactly the same elements.

1.6 Complement

2. Let $\boldsymbol{A = \{0, 2, 4, 6, 8\}}$ and $\boldsymbol{B = \{1, 3, 5, 7\}}$ have universal set $\boldsymbol{U = \{0, 1, 2, ..., 8\}}$. Find:

a) $\bar{A} = B$

b) $\bar{B} = A$

c) $A \cap \bar{A} = \emptyset$

d) $A \cup \bar{A} = U$

e) $A - \bar{A} = A$

f) $\overline{A \cup B} = \overline{U} = \emptyset$

g) $\bar{A} \cap \bar{B} = \emptyset$

h) $\overline{A \cap B} = U$

i) $\bar{A} \times B = B \times B = \{$
$(1,1), (1,3), (1,5), (1,7),$
$(3,1), (3,3), (3,5), (3,7),$
$(5,1), (5,3), (5,5), (5,7),$
$(7,1), (7,3), (7,5), (7,7)$
$\}$

1.7 Venn Diagrams

1.8 Indexed Sets

2. Suppose
$\boldsymbol{A_1 = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24\}}$,
$\boldsymbol{A_2 = \{0, 3, 6, 9, 12, 15, 18, 21, 24\}}$,
$\boldsymbol{A_3 = \{0, 4, 8, 12, 16, 20, 24\}}$.

a) $\bigcup_{i=1}^{3} A_i = \{0, 2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24\}$

b) $\bigcap_{i=1}^{3} A_i = \{0, 12, 24\}$

4. For each $\boldsymbol{n \in \mathbb{N}}$, let $\boldsymbol{A_n = \{-2n, 0, 2n\}}$.

a) $\bigcup_{i\in\mathbb{N}} A_i = \{2n : n \in \mathbb{Z}\}$

b) $\bigcap_{i\in\mathbb{N}} A_i = \{0\}$

6.

a) $\bigcup_{i\in\mathbb{N}} [0, i+1] = [0, \infty)$

b) $\bigcap_{i\in\mathbb{N}} [0, i+1] = [0, 2]$

8.

a) $\bigcup_{\alpha\in\mathbb{R}} \{\alpha\} \times [0,1] = \mathbb{R} \times [0,1] = \{(x,y) : x \in \mathbb{R},\ 0 \le y \le 1\}$

b) $\bigcap_{\alpha\in\mathbb{R}} \{\alpha\} \times [0,1] = \emptyset$

10.

a) $\bigcup_{x\in[0,1]} [x,1] \times [0,x^2] = \{(u,v) : 0 \le u \le 1,\ 0 \le v \le u^2\}$

b) $\bigcap_{x\in[0,1]} [x,1] \times [0,x^2] = \{(1,0)\}$

12. If $\bigcap_{\alpha\in I} A_\alpha = \bigcup_{\alpha\in I} A_\alpha$, what do you think can be said about the relationships between the sets A_α?

They must all be equal.

14. If $J \neq \emptyset$ and $J \subseteq I$, does it follow that $\bigcap_{\alpha\in I} A_\alpha \subseteq \bigcap_{\alpha\in J} A_\alpha$? Explain.

Yes.
Assume $J \neq \emptyset$ and $J \subseteq I$, then,
take $x \in \bigcap_{\alpha\in I} A_\alpha$, this means $x \in A_\alpha$ for every $\alpha \in I$,
since $J \subseteq I$, then every element of J is also an element of I,
thus $x \in A_\alpha$ for every $\alpha \in J$, then, by definition, $x \in \bigcap_{\alpha\in J} A_\alpha$,
so we conclude that $\bigcap_{\alpha\in I} A_\alpha \subseteq \bigcap_{\alpha\in J} A_\alpha$. $\square$